28 B Electron can excite gas to n = 4 only, since $E_4 - E_1 = -1.52 - (-5.13) = 3.61\text{ eV}$ but $E_5 - E_1 = -1.38 - (-5.13) = 3.75\text{ eV}$.

De-excitation	Energy of emitted photon	Larger than work function?
4 → 1	3.61 eV	Yes
4 → 2	1.51 eV	Yes
4 → 3	0.42 eV	No
3 → 1	3.19 eV	Yes
3 → 2	1.09 eV	No
2 → 1	2.10 eV	Yes